

PES2UG24CS021

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D-SECTION

<https://github.com/codespiderr/PES2UG24CS201-pes-vcs>

```
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
vboxuser@ubuntu22:~/Desktop/osorange/PES2UG24CS201-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

```
lcrypto.so -lcrypto
vboxuser@ubuntu22:~/Desktop/osorange/PES2UG24CS201-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
vboxuser@ubuntu22:~/Desktop/osorange/PES2UG24CS201-pes-vcs$
```

```
untracked files by PES2UG24CS201-pes-vcs$ ./pes status
vboxuser@ubuntu22:~/Desktop/osorange/PES2UG24CS201-pes-vcs$ ./pes status
Staged changes:
  staged:    file1.txt
  staged:    file2.txt

Unstaged changes:
(nothing to show)

Untracked files:
  untracked: test_sequence.sh
  untracked: object.c
  untracked: commit.c
  untracked: test_tree
  untracked: README.md
  untracked: index.h
  untracked: pes.c
  untracked: .gitignore
  untracked: tree.c
  untracked: Makefile
  untracked: tree.h
  untracked: commit.h
  untracked: index.c

vboxuser@ubuntu22:~/Desktop/osorange/PES2UG24CS201-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776455
69 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776455
69 6 file2.txt
```

```
vboxuser@ubuntu22:~/Desktop/osorange/PES2UG24CS201-pes-vcs$ ./pes log
commit 1d0ae68d82dd8c61222b96927f0c71bd948769c499fe5391cc0001ea0ae47b66
Author: Jaladi Suhas <PES2UG24CS201>
Date: 1776455172
```

```
Add farewell
```

```
commit 88ccbb4e194cbd60ab6c09b740968b5fa83651277b9c02a0e5297c64bc867c9c
Author: Jaladi Suhas <PES2UG24CS201>
Date: 1776455171
```

```
Add world
```

```
Add farewell
```

```
commit e30ff4bc80d992ad45cf6fca07cbe1eb8464d93f0ad021cc924d0dc4454988bc
Author: Jaladi Suhas <PES2UG24CS201>
Date: 1776455171
```

```
vboxuser@ubuntu22:~/Desktop/osorange/PES2UG24CS201-pes-vcs$ find .pes -type f | sort
```

```
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/1d/0ae68d82dd8c61222b96927f0c71bd948769c499fe5391cc0001ea0ae47b66
.pes/objects/2c/f8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/88/ccbb4e194cbd60ab6c09b740968b5fa83651277b9c02a0e5297c64bc867c9c
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e3/0ff4bc80d992ad45cf6fca07cbe1eb8464d93f0ad021cc924d0dc4454988bc
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
```

```
vboxuser@ubuntu22:~/Desktop/osorange/PES2UG24CS201-pes-vcs$ cat .pes/refs/heads/main
1d0ae68d82dd8c61222b96927f0c71bd948769c499fe5391cc0001ea0ae47b66
vboxuser@ubuntu22:~/Desktop/osorange/PES2UG24CS201-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
vboxuser@ubuntu22:~/Desktop/osorange/PES2UG24CS201-pes-vcs$
```

```
vboxuser@ubuntu22:~/Desktop/osorange/PES2UG24CS201-pes-vcs$ make test-integration
```

```
=== Running integration tests ===
```

```
bash test_sequence.sh
```

```
=== PES-VCS Integration Test ===
```

```
--- Repository Initialization ---
```

```
Initialized empty PES repository in .pes/
```

```
PASS: .pes/objects exists
```

```
PASS: .pes/refs/heads exists
```

```
PASS: .pes/HEAD exists
```

```
--- Staging Files ---
```

```
Status after add:
```

```
Staged changes:
```

```
  staged:    file.txt
```

```
  staged:    hello.txt
```

```
Unstaged changes:
```

```
(nothing to show)
```

```
Untracked files:
```

```
(nothing to show)
```

```
--- First Commit ---
```

```
Committed: 0c266f7e3189... Initial commit
```

```
Log after first commit:
```

```
commit 0c266f7e3189885cc0c063918ab2da4993bb1cda0a4f1c2e5a279bfa3293e0fc
```

```
Author: Jaladi Suhas <PES2UG24CS201>
```

```
Date: 1776455512
```

```
Initial commit
```

```
--- Second Commit ---
```

```
Committed: aceb76549b4b... Update file.txt
```

```
--- Third Commit ---
```

```
Committed: b3be0e44cc5d... Add farewell
```

```
--- Full History ---
```

```
commit b3be0e44cc5d01988959cecef62f6d87196dc5a38ff9e5cebdc3e1c3e0b2b9a
```

```
Author: Jaladi Suhas <PES2UG24CS201>
```

```
Date: 1776455513
```

```
-----
```

```
commit 0c266f7e3189885cc0c063918ab2da4993bb1cda0a4f1c2e5a279bfa3293e0fc
```

```
Author: Jaladi Suhas <PES2UG24CS201>
```

```
Date: 1776455512
```

```
Initial commit
```

```
--- Reference Chain ---
```

```
HEAD:
```

```
ref: refs/heads/main
```

```
refs/heads/main:
```

```
b3be0e44cc5d01988959cecef62f6d87196dc5a38ff9e5cebdc3e1c3e0b2b9a
```

```
--- Object Store ---
```

```
Objects created:
```

```
10
```

```
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
```

```
.pes/objects/0c/266f7e3189885cc0c063918ab2da4993bb1cda0a4f1c2e5a279bfa3293e0fc
```

```
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
```

```
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
```

```
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
```

```
.pes/objects/ac/eb76549b4b7bb569e8e15c3ef0ce281eebed1901916b521477917296468205
```

```
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
```

```
.pes/objects/b3/be0e44cc5d01988959cecef62f6d87196dc5a38ff9e5cebdc3e1c3e0b2b9a
```

```
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48baf9aa3829f61f75b8d10d5350f9
```

```
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
```

```
=== All integration tests completed ===
```

```
vboxuser@ubuntu22:~/Desktop/osorange/PES2UG24CS201-pes-vcs$
```

#### Q5.1

To implement checkout, we first update the .pes/HEAD file so that it points to the selected branch. Then we read the commit hash stored in that branch file and load the corresponding tree object. After that, we update the working directory so that all files match the snapshot stored in the tree. This includes creating new files, updating modified ones, and deleting files that are not present in the target commit. The main challenge is handling conflicts when local changes exist.

#### Q5.2

A working directory is considered dirty if files have been modified after staging or committing. This can be checked by comparing the current file metadata like modification time and file size with the values stored in the index. If they differ, the file is likely modified. For better accuracy, the file can be hashed again and compared with the stored hash. Also, comparing the index with the latest commit helps identify staged changes.

#### Q5.3

In a detached HEAD state, the HEAD points directly to a commit instead of a branch. Any new commits created in this state are not linked to a branch, so they can become unreachable later. However, they are not lost immediately. They can be recovered using the reflog, which tracks recent HEAD positions. A new branch can then be created pointing to that commit to make it permanent.

#### Q6.1

Garbage collection can be implemented by starting from all branch references and traversing through all commits. From each commit, we follow links to tree objects and then to blob objects, marking all reachable objects. Any object in .pes/objects that is not reachable can be safely deleted. This ensures that only useful data is stored, improving space efficiency.

#### Q6.2

If garbage collection runs at the same time as a commit, it may delete objects that are still being used by the commit process. This creates a race condition and can corrupt the repository. To avoid this, systems delay deletion of recently created objects or use locking mechanisms. This ensures that objects are not removed while they are still needed.